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Volume 01 — Networking Foundations (L01–L10)

> Ethical Hacking · NPTEL 106105217 · Prof. Indranil Sengupta
> **notes-3** — concept exam notes (what things are / what they do). No lab command dumps.

Ethical hacking intro, OSI/TCP-IP, IP addressing, TCP/UDP, subnetting — concepts only.

L1: Introduction to Ethical Hacking

What it is / what it does

- **Hacking** = creative problem-solving applied to systems; **ethical hacking** = authorized security assessment with written permission.
- **Threat actors:** script kiddies, hacktivists, insiders, organized crime, nation-states.
- **Security goals (CIA triad):** Confidentiality, Integrity, Availability (+ Authentication, Non-repudiation).
- **Penetration testing phases:** Reconnaissance → Scanning → Gaining Access → Maintaining Access → Covering Tracks → Reporting.
- **Legal framework:** scope of engagement, rules of engagement (RoE), liability waiver.

Defenses (one-liners)

- Security policy and acceptable-use agreements
- Bug bounty / responsible disclosure programs
- Defense-in-depth architecture
- Regular third-party audits

Exam lines

- Ethical hacking requires **explicit written permission**; unauthorized access is a criminal offense.

- CIA triad is the foundation of information security objectives.
- Pentest phases: recon, scan, exploit, maintain, report.
- White-hat vs black-hat vs grey-hat distinction.
- CVSS scores rate vulnerability severity (0–10).

L2: Basic Concepts of Networking — Part I

What it is / what it does

- **Network types:** LAN, MAN, WAN; client-server vs peer-to-peer.
- **OSI 7-layer model:** Physical → Data Link → Network → Transport → Session → Presentation → Application.
- **PDU names:** bits, frames, packets, segments, data.
- **Topologies:** bus, star, ring, mesh, hybrid.
- **Transmission media:** twisted pair (UTP/STP), coaxial, fiber optic, wireless.

Defenses (one-liners)

- Replace hubs with switches (microsegmentation)
- VLAN segmentation
- Physical port security (802.1X)

Exam lines

- OSI Layer 3 = Network (routing); Layer 4 = Transport (TCP/UDP).
- Star topology: single switch failure isolates nodes; mesh is most fault-tolerant.
- LAN = local; WAN spans geographic distances.
- ping uses ICMP Echo Request/Reply (Layer 3).
- PDU at Data Link layer is a **frame**.

L3: Basic Concepts of Networking — Part II

What it is / what it does

- **MAC address:** 48-bit hardware address (OUI + NIC-specific); flat namespace.
- **Ethernet frame structure:** preamble, dest/src MAC, type/length, payload, FCS.
- **CSMA/CD** (legacy Ethernet): collision detection on shared bus.
- **Switches:** learn MAC→port via CAM table; forward unicast, flood unknown/broadcast.
- **VLANs (802.1Q):** logical segmentation; tag inserted in frame header.

Defenses (one-liners)

- Port security (limit MACs per port)
- Dynamic ARP Inspection (DAI)
- VLAN access control lists

Exam lines

- Switch operates at **Layer 2**; router at **Layer 3**.
- CAM table maps MAC address → switch port.
- Broadcast MAC = FF:FF:FF:FF:FF:FF.
- 802.1Q VLAN tag is 4 bytes inserted in Ethernet frame.
- MAC flooding causes switch to flood frames to all ports (fail-open).

L4: TCP/IP Protocol Stack — Part I

What it is / what it does

- **TCP/IP model (4 layers):** Network Access → Internet → Transport → Application.
- Maps loosely to OSI: Layers 1–2 → Network Access; Layer 3 → Internet; Layer 4 → Transport; Layers 5–7 → Application.
- **Encapsulation:** each layer adds its header (and sometimes trailer) to data from layer above.
- **Internet layer protocols:** IP (v4/v6), ICMP, IGMP, ARP.
- **IP datagram:** version, header length, TTL, protocol field, src/dst address, payload.

Defenses (one-liners)

- Encrypt at Transport (TLS) or Application layer
- Network segmentation to limit broadcast domains
- Egress filtering

Exam lines

- TCP/IP has **4 layers**; OSI has **7**.
- IP operates at Internet layer; provides logical addressing only (no reliability).
- TTL field prevents infinite routing loops (decremented per hop).
- Protocol field in IP header identifies TCP (6) or UDP (17).
- Encapsulation: Application data → TCP segment → IP packet → Ethernet frame.

L5: TCP/IP Protocol Stack — Part II

What it is / what it does

- **Application layer protocols:** HTTP (80), HTTPS (443), DNS (53), SMTP (25), FTP (21), SSH (22), Telnet (23).
- **Socket API:** (IP address, port number, protocol) uniquely identifies endpoint.
- **Well-known ports:** 0–1023; registered 1024–49151; dynamic/ephemeral 49152–65535.
- **DNS resolution:** recursive (client→resolver) vs iterative (resolver→root→TLD→authoritative).
- **DHCP (DORA):** Discover, Offer, Request, Acknowledge — auto IP configuration.

Defenses (one-liners)

- DNSSEC (signed DNS responses)
- Disable zone transfers (AXFR) to unauthorized clients
- Split-horizon DNS for internal vs external views

Exam lines

- DNS uses **UDP port 53** (TCP for large transfers/zone transfers).
- DHCP provides IP, subnet mask, gateway, and DNS server.
- Socket = IP + port + protocol (TCP/UDP).
- Telnet (port 23) transmits credentials in cleartext — never use in production.

L6: IP Addressing and Routing — Part I

What it is / what it does

- **IPv4 address:** 32 bits, dotted-decimal notation (e.g., 192.168.1.10).
- **Address classes (legacy):** A (0–127), B (128–191), C (192–223); classless CIDR replaced this.
- **Private ranges (RFC 1918):** 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16.
- **Special addresses:** loopback 127.0.0.0/8, link-local 169.254.0.0/16, broadcast (host bits all 1s).
- **Static vs dynamic** addressing; NAT maps private→public.

Defenses (one-liners)

- Network Address Translation (hide internal topology)
- RFC 1918 addressing with firewall rules
- IP allowlisting for management interfaces

Exam lines

- Private ranges: 10.x, 172.16–31.x, 192.168.x (RFC 1918).
- Loopback = 127.0.0.1; used for local service testing.
- CIDR notation: IP/prefix-length (e.g., /24 = 255.255.255.0).
- NAT translates private IPs to one public IP (PAT/overloading).
- Classless addressing (CIDR) replaced classful A/B/C.

L7: IP Addressing and Routing — Part II

What it is / what it does

- **Routing table entries:** destination network, next-hop, interface, metric.
- **Default gateway:** route for all non-local destinations (0.0.0.0/0).
- **Static routing:** manually configured; predictable but inflexible.
- **Dynamic routing:** routers exchange topology via routing protocols.

- **Longest prefix match:** most specific route wins when multiple match.

Defenses (one-liners)

- Route authentication (OSPF MD5, BGP MD5/TCP-AO)
- Filter routing updates at edge
- Disable unnecessary routing protocols on endpoints

Exam lines

- Default route = 0.0.0.0/0 pointing to gateway.
- Longest prefix match selects the most specific route.
- Static routes: admin-configured; no auto-failover.
- Traceroute exploits ICMP Time Exceeded (TTL expiry) messages.
- Host uses routing table to decide local delivery vs forwarding to gateway.

L8: TCP and UDP — Part I

What it is / what it does

- **TCP:** connection-oriented, reliable, ordered delivery, flow control, congestion control.
- **Three-way handshake:** SYN → SYN-ACK → ACK (establishes connection).
- **TCP header fields:** src/dst port, seq number, ack number, flags (SYN/ACK/FIN/RST/PSH), window size.
- **UDP:** connectionless, no guarantee of delivery, no ordering, lower overhead.
- **Use cases:** TCP = HTTP, SSH, FTP; UDP = DNS, VoIP, streaming, gaming.

Defenses (one-liners)

- SYN cookies (mitigate SYN flood)
- Stateful firewall (track connection state)
- Disable unnecessary services

Exam lines

- TCP 3-way handshake: SYN (seq=x) → SYN-ACK (seq=y, ack=x+1) → ACK (ack=y+1).
- SYN scan is stealthier than connect scan (no full connection).
- UDP has no handshake; stateless and faster but unreliable.
- TCP sequence numbers enable ordered, reliable delivery.
- RST flag immediately terminates a TCP connection.

L9: TCP and UDP — Part II

What it is / what it does

- **TCP connection teardown:** FIN → ACK → FIN → ACK (four-way) or RST for abrupt close.
- **TCP state machine:** CLOSED → SYN_SENT → ESTABLISHED → FIN_WAIT → CLOSED.
- **Flow control:** sliding window, receiver advertises window size.
- **Congestion control:** slow start, congestion avoidance, fast retransmit (TCP Reno/CUBIC).
- **Port scanning types:** SYN, connect, FIN, NULL, Xmas, ACK, UDP.

Defenses (one-liners)

- IDS/IPS detecting scan patterns
- Rate limiting on SYN packets
- Close unused ports; use port knocking for admin access

Exam lines

- FIN scan: send FIN packet; closed port → RST, open port → no response (RFC-compliant).
- Filtered port: firewall drops probe — nmap reports “filtered.”
- UDP scanning is slower and less reliable than TCP scanning.
- TCP sliding window provides flow control between sender and receiver.
- Four-way teardown uses FIN and ACK flags in both directions.

L10: IP Subnetting

What it is / what it does

- **Subnet mask:** separates network prefix from host portion (e.g., /24 = 255.255.255.0).
- **Subnetting:** divide a network into smaller broadcast domains.
- **CIDR calculation:** number of hosts = 2^(32-prefix) − 2 (subtract network and broadcast).
- **VLSM:** Variable Length Subnet Mask — different subnet sizes in same network.
- **Supernetting (route aggregation):** combine contiguous subnets into one route advertisement.

Defenses (one-liners)

- Microsegmentation with /26 or smaller subnets
- Network ACLs between subnets
- Monitor ARP traffic for rogue devices

Exam lines

- /24 subnet: 256 addresses, 254 usable hosts.
- /30 subnet: 4 addresses, 2 usable (common for point-to-point links).
- Subnet mask AND IP address = network address.
- VLSM allows efficient IP allocation with different prefix lengths.